Open questions from Grothendieck's Montpellier notes: fourteen statements a formalisation could not reach

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Abstract

The University of Montpellier has put online 178 folders of Alexandre Grothendieck's mathematical working notes. Since August 2026 an edition has transcribed 165 of them with a language model, restated each in current notation, searched the literature for the statements they contain, and formalised what it could in Lean 4 against Mathlib. Most of the statements that look new sit in theories Mathlib does not have yet: étale cohomology, topoi, stacks, abelian schemes, motives. This note collects fourteen of them for mathematicians. Each comes with its folder and page, the context, what the literature search found, the argument the page gives or sketches, the steps the edition added, and what would settle it. None of the statements has been checked by a specialist, and several may already be in print; a reference is as welcome as a proof or a counterexample.

1 Why this note

Grothendieck gave his mathematical papers to Jean Malgoire in 1990 and 1995. They were deposited at the University of Montpellier in 2010 and given to it in 2012, and the university placed 178 folders, 16,074 pages by its inventory's count, in open access as facsimiles. They are working notes: drafts in ink and pencil, restarted, struck, annotated, almost never dated. Twenty folders have been edited by hand. The rest had not been transcribed.

Since August 2026 an edition has read the remaining folders with a language model as first transcriber, and written for each folder a *modernised reading*, a restatement in current notation made from the transcription and never from the hand. A register of *findings* lists the statements of the folders that may not stand in the literature, each written as a claim about the literature with what was searched and what would settle it; at the time of writing it holds 456 rows, of which 83 turned out to be in print. Statements were then formalised in Lean 4 against a pinned revision of Mathlib, about two thousand theorems in all. A companion paper describes the formalisation [16].

Lean settles a statement only when the objects it speaks of exist in the library. In October 2026, 203 open findings with no proof were sorted by what Mathlib can reach: 16 were reachable as it stands, 39 in part and 147 not at all, and one was set aside as ill-posed; the 147 are out of reach because they live in étale or crystalline cohomology, topoi, stacks, derivators, motives or Hodge theory [16]. Those are exactly the parts of the notes where Grothendieck's own work was. The aim of this note is to hand the most promising of them to people who can prove them with pen and paper, or say where they are already proved.

How to read the entries

Each problem gives its *source*: the folder (a link to Montpellier's facsimile), the pages in the archive's pagination, and the inventory's dating, whose brackets mark an inferred date. « The page » means what Grothendieck wrote, as transcribed; short quotations are his. « The reading » means the edition's modernised reading, made by a language model. « We » marks a step of the edition: an argument, a correction or a search made for this note or for the register, with a language model, and not checked by a specialist unless the entry says so. The entries make no claim of priority. A statement on an undated leaf says when he wrote it down and nothing about whether others had.

The transcriptions themselves are not public: the rights belong to Grothendieck's heirs, and the reading views stay off the site until they authorise them. The facsimiles are public, at Montpellier, and each entry cites the page to look at. Montpellier's servers were down at the end of October 2026; the links follow the archive's own scheme and should work again.

2 Étale cohomology and Hodge theory

Problem 1 (A global bound with a shift of 3)**.** For a field k of characteristic p put $p^{\nu(k)} = [k : k^p]$, and for a scheme Z of characteristic p let $\nu(Z)$ be the supremum of $\nu(k(z))$ over its maximal points. Let X be an affine scheme on which p is not invertible, so that $V(p) \subset X$ has characteristic p. Is it true that

$$\operatorname{cd}_p(X) \;\le\; \sup\Big(\sup_{x\in X} \operatorname{cd}_p k(x),\; \nu(V(p)) + 3\Big)\,?$$

Source. folder 31, p. 84, *SGA A [SGA 4] (résidus de rédaction)*, inventory dating [1963–1973]. The page is headed « Conjectures sur la cohomologie des schémas affines »; this is its conjecture 2°). The page writes ℓ for the prime; we write p, since it is the characteristic of $V(p)$ and not a prime invertible on X.

Context. M. Artin wrote in SGA 4 X 3.0 that in unequal characteristic one must « certainement tenir compte » of the rank of Ω^1_k, without a bound [34]. The same page states a local form, 4°): for U open in the spectrum of a henselian local domain A of residue characteristic p with fraction field of characteristic 0, $\operatorname{cd}_p(U) \le \dim A + \nu(k) + \operatorname{cd}_p(k)$. The words around « ouvert » in 4°) are hard to read, so whether U is meant to be affine, and whether it may meet $V(p)$, is not known.

What is known. The local form holds, in a sharper form, for U a non-empty affine open of $\operatorname{Spec} A[1/p]$ with A an excellent henselian local domain and $\operatorname{char} K = 0$: Gabber and Orgogozo prove $\operatorname{cd}_p(U) = \dim A + \dim_p(k)$ [13, Th. 6.1], and Kato's p-dimension satisfies $\dim_p(k) \le \nu(k) + \operatorname{cd}_p(k)$, which is immediate from their Définition 1.1. Mathew proves that the residue fields of $\operatorname{Spec} R[1/p]$ have $\operatorname{cd}_p \le d+1$ when R is an excellent normal domain, henselian along an ideal $I \ni p$, with $\dim R_{\mathfrak p} + \log_p[\kappa(\mathfrak p) : \kappa(\mathfrak p)^p] \le d$ for $\mathfrak p \supseteq I$; the same bound for the affine opens $\operatorname{Spec} R[1/f]$ of $\operatorname{Spec} R[1/p]$ is established in the course of the proof [24, Prop. 4.5]. Globally, SGA 4 gives $\operatorname{cd}_p X \le 2\dim X + 1$ for X of finite type over $\mathbb{Z}$ [34, X 6.2], and Clausen and Mathew give $\operatorname{cd}_p(X) \le \dim X + \sup_x \operatorname{cd}_p k(x)$ for X qcqs of finite Krull dimension [7, Cor. 3.29]; for $\operatorname{Spec} \mathbb{Z}_p[x_1, \dots, x_n]$, which is not of finite type over $\mathbb{Z}$, the latter gives $2n+3$

where the page's bound gives $n+3$. Gabber and Zavyalov settle the analogous expectation for affinoids [14, Th. 1.7]; the Travaux de Gabber treat ℓ invertible only.

Ideas. The constant cannot be lowered to 2 (our observation): for $X = \operatorname{Spec}\mathbb{Z}$ and p odd, $\nu(V(p)) = 0$ and $\sup_x \operatorname{cd}_p k(x) = \operatorname{cd}_p \mathbb{Q} = 2$, while $\operatorname{cd}_p(\operatorname{Spec}\mathbb{Z}) = 3$ [34, X 6.1]; by Artin–Verdier duality $H^3(\operatorname{Spec}\mathbb{Z}, j_!\mu_p) \cong \mathbb{Z}/p$ [25, II.3.2, 3.3]. A natural route, which is the one of Mathew's proof, is the excision sequence for $V(p) \subset X$ and its complement $X[1/p]$: for X noetherian, $V(p)$ is affine of characteristic p, so of p-cohomological dimension at most 1 [34, X 5.1] (for X not noetherian a limit argument is needed); local cohomology along $V(p)$ can be moved to the henselisation along $V(p)$ by Gabber's affine analogue of proper base change, where Gabber–Orgogozo and Mathew apply. The difficulty is to absorb the dimension terms of those local bounds into $\sup_x \operatorname{cd}_p k(x)$, which the page's bound allows, and to see where the shift 3 comes from: the page does not explain it.

Lean. Not attempted: Mathlib has no étale cohomology.

What would settle it. A printed bound of this shape for an arbitrary affine scheme of mixed characteristic, or a counterexample to the 3.

Problem 2 (Lower bound for a local domain: the exceptions)**.** Let A be a noetherian local domain with fraction field K and residue field k, and r a prime. The page states that

$$\operatorname{cd}_r(K) \;\geq\; \operatorname{cd}_r(k) + \dim A$$

with exactly two possible exceptions: (a) r is the characteristic of K; (b) $r = 2$ and some point x of $\operatorname{Spec} A$ has an orderable residue field $k(x)$ with $\operatorname{cd}_2 k(x)(\sqrt{-1})$ finite. It adds that exception (b) disappears when A is regular, and reduces to « k itself real » when A is catenary and universally japanese. Are these refinements true?

Source. folder 31, p. 22, inventory dating [1963–1973]. The margin gives the example that keeps exception (b) alive: the local ring at the origin of $\mathbb{R}[X,Y]/(X^2+Y^2)$, a real domain whose only real point is the origin (Artin's example after SGA 4 X 2.3). The margin also asks: « cette restriction est-elle essentielle ??? »

What is known. The inequality is SGA 4 X 2.4 when r is invertible in A, with the cruder exception « some residue field orderable » [34]. When r is the residue characteristic and $\operatorname{char} K = 0$, it follows for A excellent henselian from Gabber–Orgogozo §3 [13]. We found nothing on the finer exception (b), on the regular case, on the catenary refinement, or on r the residue characteristic for A neither excellent nor henselian. Serre's *Cohomologie galoisienne* II §4 and Scheiderer's *Real and Étale Cohomology* [33], where cohomological dimension in the presence of orderings is treated, remain to be read.

Ideas. Artin's proof of X 2.4 goes by induction on $\dim A$, down to a henselian local ring of dimension one through a height-one prime, where X 2.3 and 2.2 apply [34]. The finer exception (b) says that what matters is the virtual cohomological dimension of the real residue fields, which suggests running the same induction with cd_2 replaced by cd_2 of $k(x)(\sqrt{-1})$ at the real points. The page's exception list covers r equal to the residue characteristic in mixed characteristic, which SGA 4 X 2.4 does not.

Lean. Not attempted (Galois cohomology of fields is in Mathlib only in low degree).

What would settle it. A reference for (b) and the two refinements, or an example of a regular local domain where the inequality fails at $r = 2$.

Problem 3 (Hodge types of the deleted neighbourhood of a point)**.** Let Y be a complex algebraic variety of dimension n, $j : X \hookrightarrow Y$ an open immersion with X smooth and dense, $y \in Y \smallsetminus X$, and $H^i = H^i(X \cap B_y) = (R^i j_* \mathbb{Q})_y$ with its mixed Hodge structure (Durfee [10] when $(Y \smallsetminus X) \cap B_y = \{y\}$; in general through Saito's mixed Hodge modules [32]). Then

$$h^{p,q}(H^i) \neq 0 \implies p, q \in [\max(0, i-n+1),\ \min(i, n)],$$

so the weights lie in $[0, 2i]$ for $i \leq n-1$ and in $[2(i-n+1), 2n]$ for $n \leq i \leq 2n-1$ (and $H^i = 0$ for $i \geq 2n$), and the level $\max |p-q|$ is at most $\min(i, 2n-1-i) \leq \min(i, n-1)$. Is this in the literature, and what is the right relative statement for a proper morphism?

Source. folder 12, pp. 115–119, *Généralités (sous-groupes de Galois motiviques)*, inventory dating 1964–[c. 1971]. Page 119 boxes « poids $\leq 2\inf(i, n)$, niveau $\leq \inf(i, n-1)$ » for X regular, $F = \mathbb{Q}_\ell$ and f an immersion with $n = \dim \overline{f(X)}$, in his yoga of weights and levels of ℓ-adic sheaves. The same page asks, with a question mark, for a relative version with $n = d_y(f)$ and level $\leq \min(i, n)$. The typescript says of its own conclusions (p. 96): « la portée des « résultats » de notre travail est nécessairement sujette à caution, puisqu'ils dépendent de conjectures non prouvées à l'heure actuelle » (the Weil conjectures and resolution of singularities).

Context. The page's argument is a Leray spectral sequence for a factorisation of f as an open immersion with normal crossings complement followed by a proper map, conditional on resolution, purity and the Weil conjectures, with tables of weights and levels. The Hodge-theoretic statement above is our translation.

What is known. Deligne bounds the Hodge types of the cohomology of a variety [9, 8.2.4]; Durfee constructs the mixed Hodge structure on $H^i(X \cap B_y)$ and its weight spectral sequence; Durfee–Saito, Steenbrink, Navarro Aznar and El Zein bound weights; for isolated singularities the box is implicit in the literature on links: it follows from semipurity and duality as in [12, Th. 2.6, Def. 2.7]. We did not find the box, or the level bound, for arbitrary y.

Ideas. Our argument: take a resolution $\pi : \tilde{Y} \to Y$, an isomorphism over X, with $D = \pi^{-1}(Y \smallsetminus X)$ simple normal crossings; put $F = \pi^{-1}(y)$ and $F_q = F \cap D^{(q)}$. Then $\dim F_0 \leq n-1$ and $\dim F_q \leq n-q$. The weight spectral sequence $E_1 = H^{k-q}(F_q)(-q) \Rightarrow H^k(X \cap B_y)$ (Durfee's, in the case he treats; we have not checked its form in the general case) and Hodge III 8.2.4 (i), (ii) for the compact F_q put every E_1 term in the box. Locally the lower end moves up by one because $\pi^{-1}(y)$ has dimension at most $n-1$. The cone over a smooth projective V of dimension $n-1$ with $h^{i,0}(V) \neq 0$ attains level i for $i \leq n-1$ and $n-1$ at $i = n$. For a proper f with fibre dimension d the gain fails, since a smooth family of curves already has H^1 of level $1 = d$; so the relative question of page 119 cannot keep $n-1$ if $d_y(f)$ is the fibre dimension.

Lean. Not attempted: no Hodge structures on cohomology in Mathlib.

What would settle it. A reference for the box; a relative statement interpolating between the

immersion case and the smooth proper case; its ℓ-adic form with level read as divisibility of Frobenius eigenvalues.

Problem 4 (An Ext^2 that vanishes when 2 is invertible: beyond smooth coefficients)**.** On the fppf site of a base on which 2 is invertible, is $\mathcal{E}xt^2_{\mathbb{Z}}(G, M) = 0$ for every finite locally free commutative group scheme G and every quasi-coherent M, not only for the smooth coefficients covered by Breen's theorem below?

Source. folder 129, p. 95, *Notes séminaire Montréal*, inventory dating [1970]. A parenthetical NB after a theorem on Ext over truncated rings: « $\mathrm{Ext}^2_{\mathbb{Z}}(G, M) = 0$ si 2 inv., pour tt G fini loc. libre », with the remark that this simplifies the deformation theory of Barsotti–Tate groups for $p \neq 2$. The word « inv. » is uncertain in the transcription, and the page gives no proof.

What is known. Breen proved, five years later, that $\mathcal{E}xt^i(G, H) = 0$ for $1 < i < 2p - 2$, for G finite flat and H smooth commutative quasi-projective, when every residue characteristic is at least p and multiplication by every prime $q < p$ is an automorphism of G; and that $\mathcal{E}xt^{2p-2}(\alpha_p, H) \neq 0$ in characteristic p for $H = \mathbb{G}_a, \mathbb{G}_m$ [4]. With $p = 3$ this gives the NB for smooth coefficients, in particular for vector bundles, when 2 is invertible; with $p = 2$ it shows $\mathcal{E}xt^2(\alpha_2, \mathbb{G}_a) \neq 0$, which supports the reading « si 2 inv. ». Illusie places the obstruction to deforming a truncated Barsotti–Tate group over a perfect field in $\mathrm{Ext}^2_{\mathbb{Z}}(G_0, t_{G_0} \otimes J)$ and says that proving directly that it vanishes is out of reach [17, p. 2]; Breen's theorem, which he does not cite, bears on that group.

Ideas. Breen's theorem is stated for the étale sheafification of the Ext sheaves. The remaining cases are a quasi-coherent M that is not locally free, and the passage from Breen's sheaves to the global groups on the fppf site, through the local-to-global spectral sequence.

Lean. Not attempted: no fppf cohomology in Mathlib.

What would settle it. Breen's vanishing for an arbitrary quasi-coherent M, or a counterexample; and whether it gives Illusie's obstruction directly.

3 Algebraic cycles

Problem 5 (Splitting the critical Lefschetz condition)**.** Let X be smooth projective of dimension n over an algebraically closed field, with a Weil cohomology H, a smooth hyperplane section $\varphi : Y \hookrightarrow X$, and L the Lefschetz operator. Write H^{2i} for the cohomology in which codimension-i cycles live, and consider:

- $A'(X)$: for $2i \leq n-1$, a class $x \in H^{2i}(X)$ whose restriction $\varphi^* x$ is algebraic is algebraic;
- $A''(X)$: for $2i \geq n-1$, every algebraic class of $H^{2i+2}(X)$ is $\varphi_* y$ for an algebraic $y \in H^{2i}(Y)$;

and their critical versions A'^0, A''^0, with $2i = n-2$ for n even and $2i = n-1$ for n odd. Let $A^0(X)$ be the critical Lefschetz condition: $L^2 : H^{n-2}(X) \to H^{n+2}(X)$ for n even, $L : H^{n-1}(X) \to H^{n+1}(X)$ for n odd, is a bijection on algebraic classes. Under which hypotheses do $A'^0(X)$ and $A''^0(X)$ imply $A^0(X)$, and is the resulting reduction of

the Lefschetz standard conjecture,

$$C(X) \iff C(Y) + A'^0(X \times X) + A''^0(X \times X),$$

correct as stated?

Source. folder 16, p. 49, *Formalisme algébrique des correspondances et algèbres de Poincaré-Hodge*, inventory dating 1966–1967, in a letter of January 1967. The letters use A, C, D in senses that do not match the later published labels; the conditions above are the reading's restatement. The letter to Coates places the question in the motivic picture (p. 48): « The complete yoga then being that we have the category of motives which is semi-simple ! ».

What is known. The equivalence $C(X) \iff C(Y) + A^0(X \times X)$ and its descent along a chain of hyperplane sections are Kleiman's [20, Th. 2.13]. The splitting of A^0 into the two one-sided conditions was not found in Kleiman 1968 and 1994, Grothendieck 1969, Charles 2013, Ramón Marí, or Milne; André's book [1, ch. 5, §5.2] remains to be read.

Ideas. The page factors the critical operator through the section. For $n = 2m$: $L^2 = \varphi_* \circ L_Y \circ \varphi^* : H^{2m-2}(X) \to H^{2m-2}(Y) \to H^{2m}(Y) \to H^{2m+2}(X)$; for $n = 2m+1$: $L = \varphi_*\varphi^* : H^{2m}(X) \to H^{2m+2}(X)$. With it (our reconstruction, $n = 2m$): an algebraic $z \in H^{2m+2}(X)$ is $\varphi_* y$ with y algebraic by A''^0; by weak and hard Lefschetz, $y = L_Y \varphi^* x$ for a unique x, and $z = L^2 x$; if L_Y^{-1} preserves algebraic classes on Y, $\varphi^* x$ is algebraic, and then x is algebraic by A'^0. So the splitting holds given the Lefschetz condition on the section in the relevant degree. In the displayed equivalence it is applied to $T = X \times X$, whose hyperplane section is not $Y \times Y$; whether the hypothesis on the section is supplied by the induction, or must be added, is the point to check.

Lean. Not attempted: no Weil cohomology in Mathlib.

What would settle it. A reference for the splitting, or a precise statement of the hypotheses under which it holds, applied to $X \times X$.

4 Abelian varieties and reductive groups

Problem 6 (Polarised abelian surfaces with even Euler characteristic)**.** Let A/S be an abelian scheme of relative dimension 2 with dual B, and δ a section of $\mathrm{NS}_{A/S}$ with $\chi(\delta)$ even. Is the torsor $\mathrm{Pic}^{\delta}_{A/S}$ under B trivial? Over a field k, this says that the Poonen–Stoll class $c_\delta \in H^1(k, B)$ vanishes [29].

Source. folder 59, pp. 11–13, *Construction des faisceaux inversibles sur schémas de Picard*, inventory dating [années 1960–1970]. Page 11 gives a canonical isomorphism $\mathrm{Pic}^{\delta'}_{B/S} \simeq \tilde{\delta}'_* \mathrm{Pic}^{\delta}_{A/S}$; page 13 composes it with its analogue and concludes, for $n = 2$ and $\chi(\delta)$ even, that Pic^{δ} is trivial. That last line rests on an uncertain « si » and an illegible word.

What is known. Poonen and Stoll define c_λ and prove $2c_\lambda = 0$, with no criterion in terms of $\chi(\lambda)$ or the dimension [29, §4]. We found no such criterion elsewhere.

Ideas. The reading thought the conclusion depends on compatibilities of trivialisations that the folder asks for and never proves. Our argument, unchecked, needs none for mere triviality. The map $L \mapsto \det(\hat{L})^{\pm 1}$, with $\hat{L}$ the Fourier–Mukai transform, is a morphism of fppf sheaves $\mathrm{Pic}^{\delta}_{A/S} \to \mathrm{Pic}^{\varphi(\delta)}_{B/S}$, equivariant along $\varphi(\delta)^{\sim}$ by the theorem of the square. The same construction on B goes back to $\mathrm{Pic}^{\varphi\varphi(\delta)}_{A/S}$, and in dimension 2, $\varphi\varphi(\delta) = \delta$. The composite $B \to B$ is multiplication by $\mp\chi(\delta)$. Hence $(\chi(\delta) \pm 1)\,[\mathrm{Pic}^{\delta}] = 0$, and $2\,[\mathrm{Pic}^{\delta}] = 0$ through $L \mapsto L \otimes [-1]^{*}L$. Both $\chi \pm 1$ are odd when χ is even, so the class vanishes whatever the signs. Whether the trivialisation obtained is canonical is a separate question.

Lean. Not attempted: no abelian schemes in Mathlib.

What would settle it. A specialist's check of the argument. One abelian surface over a number field with a polarisation of type $(1, 2)$ and $c_\lambda \neq 0$ refutes the statement. A sharp consequence to test (our heuristic, unchecked): for a genus-2 curve C/k whose Jacobian has real multiplication by $\mathcal{O}_D$, $D \equiv 1 \bmod 8$, defined over k, the theta class is a sum of two classes of even χ, so the statement would force Pic^{1}_{C} to have a k-point.

Problem 7 (Kostant's tori in apposition: are the Borels opposite?)**.** Let G be a complex simple adjoint group with Coxeter number h and Weyl group W, and let T, T' be maximal tori in apposition in Kostant's sense [23], with Borel subgroups $B \supset T$ and $B' \supset T'$. Are B and B' always in general position, that is, is $B \cap B'$ a maximal torus? If not for all such quadruples (T, B, T', B'), is there a distinguished conjugacy class of quadruples, stable under the outer automorphisms?

Source. folder 123, pp. 21–22, *[Autour de Kostant]*, letter to Kostant of 22 October 1969: « is it true that for some, or for all, such quadruples, B and B' are « in general position » […] ? Is there in any sense a *distinguished* conjugacy class of such quadruples […] ? » The letter states without proof that the quadruples form $|W|^2/(h\,z\,\varphi(h))$ conjugacy classes, z the order of the centre of the simply connected cover, and (in a footnote in his hand) that $N(T) \cap N(T')$ is an extension of $(\mathbb{Z}/h)^*$ by $\mathbb{Z}/h \times Z(\tilde{G})$.

What is known. The count, which this note listed as open until 26 October 2026, follows from standard theory (our argument, unchecked by a specialist). Let $x = \rho^{\vee}(e^{2\pi i/h})$, the principal element of T, and T' the centraliser of a cyclic element $E' = \sum_i c_i e_{\alpha_i} + c_0 e_{-\theta}$. An element $t \in T$ normalising T' sends E' into $\mathfrak{h}' \cap \mathfrak{g}^{\mathrm{Ad}(x)=\zeta}$, which is the line $\mathbb{C}E'$, so $\alpha_i(t) = \theta(t)^{-1} = \lambda$ with $\lambda^h = 1$: $T \cap N(T') = \langle x \rangle$, of order h. The image of $N(T) \cap N(T')$ in W is the set of w with $w(x) \in \langle x \rangle$: the stabiliser of x is the group $\Omega \cong P^{\vee}/Q^{\vee}$ of order z (in the adjoint group x has a disconnected centraliser), x is conjugate to x^k exactly for k prime to h [23, 35], and each such w lifts into $N(T')$, because regular semisimple elements of the eigenspaces $\mathfrak{g}^{\zeta^j}$, j prime to h, have the same centralisers as those of $\mathfrak{g}^{\zeta}$ and their Cartan subspaces are conjugate under $Z_G(x)^{\circ} = T$. Hence $|N(T) \cap N(T')| = h\,z\,\varphi(h)$; the stabiliser of a quadruple is $T \cap T' = 1$, and the count follows. A computer count over $\mathbf{F}_q$ (`computations/kostant`) agrees: 2, 4 and 12 orbits for A_2, B_2 and G_2. For general position we found nothing.

Ideas. For A_2 all 36 pairs (B, B') are opposite: the eigenvectors of E' are the Fourier vectors $(1, \omega^k, \omega^{2k})$, and the standard flag is transverse to every flag they span. In general the question is the Bruhat cell of (B, B') for B' running over the $|W|$ Borels through T': whether it is always

the open cell.

Lean. Not attempted in Lean. The computer count above can be extended to the relative position of B and B' in B_2 and G_2, which we have not done.

What would settle it. The relative position of B and B' for all quadruples, by an argument or by computation in each type.

5 Topoi and stacks

Problem 8 (Two classifying topoi that are not coherent)**.** The theories of commutative rings with irreducible spectrum (nilradical prime) and of rings with zero-dimensional spectrum are geometric. The page says their classifying topoi, subtopoi of the coherent classifying topos of rings, are not coherent, their presenting topologies on affine schemes of finite type over $\mathbb{Z}$ not being quasi-compact. Is that right, and do these topoi have enough points?

Source. folder 161-2, pp. 39, 41–42, *Algèbre universelle*, inventory dating [c. 1963–1973]. Page 39: « Cette topologie n'est pas quasi-compacte »; page 42, margin: « topologie non cohérente ! A-t-elle assez de pts ?? »; page 41, margin: « encore un topos non cohérent (bien que sous-topos d'un topos cohérent) ».

What is known. Locally coherent toposes have enough points (Deligne, [34, VI, 9.0]); these theories are geometric but, as presented, use the infinite disjunction hidden in « nilpotent ». We found neither non-coherence nor the existence of points stated for them, in Johnstone [18] or in Caramello's criteria for De Morgan classifying toposes, which take quotients of the theory of rings as examples [6, §5].

Ideas. Non-coherence needs more than the presentation: one must show that no coherent axiomatisation exists, for instance by exhibiting a cover in the classifying topos with no finite subcover. For points, Barr's theorem gives a Boolean cover, not points; a model-theoretic argument, or an explicit set of local rings detecting the theory, is needed.

Lean. Not attempted: classifying topoi are not in Mathlib.

What would settle it. A proof or reference for non-coherence, and an answer to the question about points.

Problem 9 (Foliations and their quotient topoi)**.** Is there an equivalence between the category of pairs (X, R), with X a topological space and R a local equivalence relation on X, locally open with locally connected fibres, and the category of triples $(X, \mathcal{Y}, f)$, with $f : \mathrm{Top}(X) \to \mathcal{Y}$ a locally open geometric morphism with locally connected fibres to a « multiplicité topologique », a 2-category whose Hom-categories are discrete?

Source. folder 119, pp. 144–145, dated « Mars 1986 » on p. 144. Under « Th »: « On a $C \rightleftarrows C'$ », both arrows marked $\approx$. No proof. A third condition on f is partly illegible; the reading suggests it may be a simple connectedness of the fibres.

What is known. SGA 4 IV, Exercise 9.8.3 j)–k), gives the functor $(X, R) \mapsto (\mathrm{Top}(X) \to \mathrm{Top}(X/R))$ [34]; Rosenthal states that the topos $\mathrm{sh}(X; r)$ exists for strictly open r (Th. 3.9, proof in his thesis) and proves it is an étendue for strictly regular r [31]; Kock and Moerdijk show that every étendue is of the form $\mathrm{sh}(M, r)$, construct the inverse $a \mapsto \mathrm{Lker}(a)$ and show the round trips are identities under simple connectedness or kernel-pair conditions [21], and study monodromy versus holonomy [22]. We found no statement of an equivalence of categories.

Ideas. The two functors are there; the problem is the right conditions on both sides so that they are mutually inverse, and which quotient topos the page means: the topos of R-invariant sheaves (sheaves on a monodromy groupoid) or the holonomy quotient, which Kock and Moerdijk distinguish.

Lean. Not attempted: no geometric morphisms of topoi in Mathlib.

What would settle it. A correct statement with proof, or Moerdijk's Zaragoza lectures [26], which we could not read and where it might be.

Problem 10 (Realising the symmetry invariant of a Picard stack)**.** Let M, N be abelian sheaves on a topos, and ${}_2N$ the 2-torsion of N. Is every homomorphism $M \to {}_2N$ the symmetry invariant of a Picard stack pinned by M and N?

Source. folder 134-1, p. 5, letter to Deligne, Lodève, 27 August 1974 (Breen's copy, forwarded by Deligne): « Je sais prouver (sauf erreur) que tout hom. $M \to {}_2N$ provient d'un champ de Picard convenable (épinglé par M, N) (a priori l'obstruction est dans $\mathrm{Ext}^3(X; M, N)$, mais un argument « universel » prouve qu'elle est nulle) ». No argument is written. On p. 6 the letter asks his own version of this problem: « Question : Le triangle exact (T) et la suite exacte (*) sont-ils connus par les compétences (Quillen, Breen, Illusie…) ? Connaissent-ils des variantes « supérieures » ? » (are they known to the experts, and do they know higher variants?).

What is known. Strict Picard stacks are classified by Ext^2 (Deligne, [34, XVIII, 1.4]); Breen classifies Gr-stacks by H^3 and gives the universal coefficient spectral sequence $\mathrm{Ext}^p(H_q(B), A) \Rightarrow H^{p+q}(B, A)$ [5, §6]. No printed source we found states the realisation, or refutes it on a topos.

Ideas. Two sketches disagree. The reading realises the invariant by hand for M free, on the split stack $M \times BN$ with an antisymmetric form b such that $b(e_i, e_i) = q(e_i)$, and passes to general M through a resolution. We doubt that step, which we have not checked: the construction depends on a basis. On the classifying topos of abelian groups, a Picard stack pinned by the generic group U and $U/2U$, with the projection as invariant, would make Moore spectra truncated above π_1 functorial in finitely presented abelian groups; but the stable Moore-space category on finitely generated groups is a non-split linear extension, split on cyclic groups [2, Rem. 1.5.5]. If that translation is right, the obstruction does not vanish universally. The statement does hold over a point and for M constant and free.

Lean. Not attempted: no stacks on a topos in Mathlib.

What would settle it. The differential $\mathrm{Hom}(M/2, N) \to \mathrm{Ext}^3(M, N)$ of the universal coefficient spectral sequence on the classifying topos of abelian groups, or Grothendieck's

« universal » argument written out.

Problem 11 (Morphisms of ramification domains are fully faithful)**.** Over a scheme S, a « domaine de ramification » is a stack of multigaloisian categories over the étale site of S with a continuous geometric realisation into relative schemes, subject to axioms R1 and R2 (pp. 190–193). For a domain (R, r) defined by a scheme Z with a finite group G of operators, is $\mathrm{Hom}((R, r), (R', r'))$ equivalent to the category of pairs (P', α), P' a galoisian object of R'_S of group G and α a G-isomorphism $r'(P') \simeq Z$, so that every homomorphism of domains is fully faithful?

Source. folder 29, p. 197, *Groupe fondamental [autour de SGA 4 et SGA 7]*, inventory dating 1959–1969: Proposition 6, stated without proof, with the corollary « nécessairement pleinement fidèle ». The axiom R2 of pp. 190–193 is itself not verified by the edition. Earlier in the folder (p. 24) he notes where removing the auxiliary choices leads: « il semble qu'on soit infailliblement conduit à prendre comme catégorie […] la catégorie […] elle-même qu'on se propose de construire et d'étudier ! », comparing it with sites and their topoi.

What is known. Nothing found. Tonini's work on stacks of ramified Galois covers [36, 37] is where an equivalent statement would sit; neither surfaced in our search, which is not the same as their not containing it.

Ideas. The corollary says that a ramification domain maps into another only as a subdomain. The proof should reduce to the case (R, r) given by (Z, G), where a morphism is determined by the image of the universal G-torsor.

Lean. Not attempted: no étale site or stacks of categories in Mathlib.

What would settle it. A proof of Proposition 6 from R1–R2, or a reference.

6 Moduli of curves and Galois actions

Problem 12 (The Galois image, seen on one curve)**.** Let X be an anabelian curve over a field K of finite type over $\mathbb{Q}$, and $\hat{\Pi}_1(X, I)$ its profinite fundamental groupoid on a finite set I of base points. The page lists necessary conditions for a vertex-fixing automorphism Θ of $\hat{\Pi}_1(X, I)$ to come from $\mathrm{Gal}(\bar{K}/K)$: it lifts to $\hat{\Pi}_1(X', I')$ for every finite étale cover X' of X defined over K; it preserves the kernels of $\hat{\pi}_1(X', x') \to \hat{\pi}_1(Y, f(x'))$ for every K-morphism $f : X' \to Y$; it is compatible with morphisms between covers; and it satisfies the loop condition at the cusps with a single multiplier. Are these conditions sufficient?

Source. folder 142, pp. 94–95, 98–101, *Action de Galois sur Teichmüller*, inventory dating [from 1978]. Page 95 conjectures the conditions sufficient « lorsque X est (disons) une courbe algébrique anabélienne, avec K extension de type fini de $\mathbb{Q}$ »; page 98 calls the loop condition « sûrement la plus importante de toutes, et de très loin ! ». Pages 98–99 are largely illegible.

What is known. Ihara's question and the Oda–Matsumoto conjecture characterise the Galois group as the automorphisms of the geometric fundamental group functor on all varieties compatible with all morphisms; Pop proved it in that form, and finite tripod variants, built

from varieties of dimension greater than one with finitely many dominant rational maps to $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ [30, Def. 2.2, Th. 2.9]. Hoshi–Mochizuki and others give injectivity for a single hyperbolic curve; Collas proposes a combinatorial model GT^{NF} as a candidate for $G_{\mathbb{Q}}$, left to forthcoming work [8, §4.3.3]. We found no characterisation attached to one curve and its covers, with base points and the inertia condition.

Ideas. Pop's variants need a base of dimension greater than one, so they do not cover the covers of one curve; that is what the page's loop condition at the cusps would have to supply. Here « anabelian curve » is the page's word for a hyperbolic curve.

Lean. Not attempted: no étale fundamental group in Mathlib.

What would settle it. A proof of sufficiency, a counterexample, or a published variant that implies it.

Problem 13 (The deleted neighbourhood of a normal crossings divisor)**.** Let X be a smooth complex analytic space (or orbifold) with a normal crossings divisor Θ, and $V^*_{\Theta,X}$ the deleted tubular neighbourhood of Θ. Is $V^*_{\Theta,X}$ the homotopy colimit of the semi-simplicial system of deleted tubular neighbourhoods of the strata of the unfolding of Θ, so that its fundamental groupoid is the amalgamated sum $\tilde{\Pi}_1 \leftarrow \tilde{\Pi}_{2,1} \rightarrow \tilde{\Pi}_2$ of the groupoids of the torus bundles over the open strata of codimension 1 and 2? And hence the same for $\Pi_1(X \setminus \Theta)$ whenever $X \setminus \Theta$ and its end have equivalent fundamental groupoids?

Source. folder 147, pp. 148–151, *Structure à l'infini des* $M_{g,\nu}$. The folder is undated; the inventory dates its group (141–149) [from 1978]–1983. We date it to the winter of 1982–1983, because its formalism is the one Grothendieck describes to Illusie in a letter of 22–25 January 1983 (IHÉS, fonds Grothendieck–Illusie, 1ILL35, p. 27) and in a *rapport d'activité* of which he sent Illusie a copy on 29 March 1983 (1ILL41, p. 4; copy in folder 119, pp. 141–142). Page 149 states a « Th. de recollement » as an equivalence of topoi and corrects it at once: « en fait, c'est une équivalence d'homotopie ». Pages 150–151 give the corollaries, with a purity argument for neglecting the strata of codimension 3. No proof is written.

Context. The folder describes the boundary of $\overline{M}_{g,\nu}$ by stable graphs, lists the strata of modular dimension at most 2, and writes on its p. 29 (archive p. 59) that a more or less complete understanding of the discrete π_1 in these small cases « donnera la clef pour une compréhension complète dans le cas général »: the two-level principle of the Teichmüller tower, about a year before the *Esquisse d'un programme*. Applied to $X \setminus \Theta = M_{g,\nu}$, the corollaries would express the mapping class group through the strata of codimension 1 and 2; that application is the reading's, not the page's.

What is known. The two-level principle for the discrete tower was proved by Hatcher, Lochak and Schneps [15]. For the general gluing statement, the places to look are the real oriented blow-up and the Kato–Nakayama space of the log structure, whose boundary is a torus-bundle model of the deleted neighbourhood, and Looijenga's and Boggi's work on the boundary of $\overline{M}_{g,n}$. We did not find the statement in this form.

Ideas. The purity step reads: removing a subset of real codimension at least 3 does not change π_1. For $M_{g,\nu}$, the hypothesis of the last corollary (end and interior with the same Π_1) should follow from the simple connectivity of the curve complex (Harer).

Lean. Not attempted: no homotopy colimits of spaces or stratified spaces in Mathlib.

What would settle it. A reference for the gluing theorem, or a proof, and the check of the hypothesis for $M_{g,\nu}$.

7 Pseudoline arrangements

A pseudoline arrangement in the real projective plane is *bigonal along* one of its pseudolines D when D carries exactly two vertices of the arrangement, so that every other pseudoline meets D at one of them. In the page's notation, II_4 is four pseudolines in general position, II_5 five in general position, $\mathrm{II}'_{3,2}$ a pencil of three and a pencil of two without the line joining their centres, and III_6 the complete quadrangle (the six lines joining four points, no three collinear).

An arrangement is *bigon-free* when none of its pseudolines is bigonal, and four pseudolines are in *general position* when no three of them pass through one point. The page's boxed dichotomy, which this note listed as an open problem until 26 October 2026, has a short proof, and so does the numbering lemma of page 28; we record it because the problem below depends on it.

Proposition (Grothendieck, folder 80, pp. 20, 24, 28; the proof is ours)**.** *A bigon-free arrangement $\mathcal{A}$ either is II_4 or contains II_5, $\mathrm{II}'_{3,2}$ or III_6; and its pseudolines can be numbered, starting from a II_4 or a III_6, so that every partial arrangement is bigon-free.*

Proof. Four pseudolines in general position. $\mathcal{A}$ is not a near-pencil, since each pencil line of a near-pencil carries only the centre and its point on the transversal; so it has a simple crossing $p = a \cap b$ [19, Th. 3.6], see also [3, Th. 1.3]. Since a is not bigonal it carries vertices $q_1, \dots, q_k$ besides p, with $k \geq 2$. Take $c \neq a$ through q_i and $d \neq a$ through q_j, $i \neq j$. Then $\{a, b, c, d\}$ is in general position as soon as $c \cap b \neq d \cap b$: indeed c, d miss p, meet a at different points, and $c \cap d$ lies neither on a nor on b. If this fails for every choice, all these pseudolines meet b at one point r; two pseudolines meet only once, so each q_i lies on a and on a single other pseudoline c_i, all through r, and every pseudoline meets a at p or at some q_i. Then $\mathcal{A} = \{a, b, c_1, \dots, c_k\}$ and b carries only p and r, so it is bigonal, which is excluded.

The dichotomy. Let $Q \subset \mathcal{A}$ be four pseudolines in general position, and $\ell \in \mathcal{A} \smallsetminus Q$. If ℓ passes through none of the six vertices of Q, then $Q \cup \{\ell\}$ is a II_5; if through exactly one, a $\mathrm{II}'_{3,2}$. Otherwise ℓ passes through exactly two, which must be an opposite pair $\{x \cap y, z \cap w\}$ of the quadrilateral $Q = \{x, y, z, w\}$ (two vertices on a common side would force ℓ to be that side). Call such an ℓ a diagonal of Q; there are at most three. If $\mathcal{A}$ contains neither II_5 nor $\mathrm{II}'_{3,2}$, then $\mathcal{A}$ is Q plus some diagonals. One diagonal alone would be bigonal; two diagonals together with Q form a III_6 (the four triple points are the vertices of the quadrangle and the six pseudolines its sides); with three, $\mathcal{A}$ contains a III_6 again.

The numbering. Adding pseudolines never removes a vertex from a pseudoline already there, so it suffices that each pseudoline, when added, meets the earlier ones in at least three points. A pseudoline meets the six sides of a III_6 in at least three points: its vertices have multiplicity at most three, and two triple points share a side, so two vertices cover at most five sides. So if $\mathcal{A} \supseteq \mathrm{III}_6$, start there and add the rest in any order. Otherwise $\mathcal{A}$ contains at most one diagonal δ of Q: start from Q, add every pseudoline other than δ (each meets Q in at least three points), and add δ last, when it has all its vertices in $\mathcal{A}$, at least three. □

This agrees with a computer enumeration of all arrangements of at most nine pseudolines (`computations/bigon`). The dichotomy is also checked in Lean for combinatorial maps (generalised maps of Euler characteristic 1 whose flags are labelled by the pseudolines; `lean/Grothendieck/Folder80Maps2.lean`, `map_dichotomy`); the numbering and the passage from a topological arrangement to its map are not formalised. We have not found the statement in print; it is elementary, and probably folklore among people who work on arrangements.

Problem 14 (A bigon-free arrangement is determined by its skeleton)**.** Is a combinatorial arrangement of pseudolines in the real projective plane without bigonal pseudolines determined by its 1-skeleton together with the decomposition of the edges into pseudolines, functorially: is the functor from bigon-free arrangements to their decomposed skeleta fully faithful, so that every automorphism of the decomposed skeleton comes from the arrangement?

Source. folder 80, pp. 39–54. The statement of page 54 (« En résumé ») is clean. It composes a full faithfulness theorem (p. 39) for arrangements with an orientation of every non-concurrent triple (pp. 41–49, conditions (A1)–(A4)) with a canonical reconstruction of these orientations from the skeleton (pp. 49–54, (A5), (A6), with the complete quadrangle as residual case). Conditions (A1)–(A6) are said to hold « bien entendu » and not proved.

What is known. Line arrangements in the projective plane are determined up to isomorphism by their circular sequences up to local equivalence (Goodman–Pollack), and the graph of a simple projective arrangement of at least four pseudolines is 4-connected [11, Th. 5.2.2, 5.5.1]. Embeddings of graphs in the projective plane are unique at face-width at least 3 and are classified at face-width 2 and 3 [38, 27, 28]. We found no statement for the projective, bigon-free case with automorphisms.

Ideas. Our pointer, unchecked: if the graph is 3-connected and its embedding has face-width at least 3, these uniqueness theorems give the statement, without even the decomposition into pseudolines, since an automorphism then maps facial walks to facial walks. A non-contractible curve meets every pseudoline, so face-width 2 means two points covering all of them; for a bigon-free arrangement this happens exactly for two pencils with no pseudoline through both centres, such as II_4 and $\mathrm{II}'_{3,2}$, which then need a separate argument, as does 3-connectivity in the non-simple case.

Lean. Partly proved, in the combinatorial maps of page 60 with its axiom that each pseudoline goes straight through each vertex (`lean/Grothendieck/Folder80Maps3.lean`). The functor is faithful (`faithful`). A skeleton isomorphism comes from an isomorphism of arrangements exactly when it preserves the face paths of length three, which is page 41's theorem (`lift_iff`), so fullness comes down to reading these paths on the skeleton. Also proved: the mod 2 homology of the map (the cycles modulo the boundaries have dimension at most 1), and (A5), the orientation of a triangle read on the skeleton when a fourth pseudoline avoids its vertices. Not formalised: (A1)–(A4), (A6) and the complete quadrangle, hence not full faithfulness.

What would settle it. A proof of (A1)–(A4), (A6), or the face-width bound with a uniqueness theorem.

8 How the list was made, and its limits

The fourteen problems were chosen from the register of findings among the rows still open and outside what Lean reached, and with a statement precise enough to be proved or refuted. Rows whose only interest is a slip on the page, a dating, or a statement now known to be in print were left out, as were the conjectures already in the published record (the standard conjectures, the section conjecture, the p-curvature conjecture, the homotopy hypothesis), which the companion paper discusses [16].

Three limits apply to every entry. The transcriptions are first passes by a language model, and doubtful readings are marked but not all checked against the facsimile; a statement may rest on a misread word, and the entries say where we know this. The literature searches were made by the same means, with full texts where they were open and abstracts otherwise; an entry that says « not found » reports a search, not an absence. And the steps marked « ours » have not been checked by a specialist.

Corrections, references and proofs are welcome.

Changes

26 October 2026: the pseudoline dichotomy of folder 80 proved (§7); the count of Kostant quadruples derived, and Problem 7 replaced by the question Grothendieck asked Kostant; Problem 4 narrowed after Breen's 1975 theorem; references corrected after a citation check.

Declaration

This note was drafted with a language model (Claude Opus 5.5, Anthropic) from the edition's registers: the findings, the modernised readings and the Lean files. The transcriptions and the modernised readings it relies on were also made with language models (Claude Opus 5 and Opus 5.5). The author received no funding for this work.

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